

Project Design Phase

Problem – Solution Fit

Date	05 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

Problem - Solution Fit Canvas — Electricity Consumption Analytics Platform

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS State electricity board grid planners & regional load dispatchers; policy analysts at central energy ministries planning renewable infrastructure.	6. CUSTOMER CONSTRAINTS CC Small data-engineering teams, limited software budgets, restricted network access beyond email/Excel, reliance on legacy on-prem systems.	5. AVAILABLE SOLUTIONS AS Manual Excel pivot tables & static PDF reports; ad-hoc requests to a central data team; generic BI tools (Power BI/Tableau) not tuned for power data.	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Forecast next-day/next-week demand per state; compare consumption across regions; prioritize renewable investment; alert before overload.	9. PROBLEM ROOT CAUSE RC Raw consumption data arrives as flat per-state, per-day CSV rows with no built-in aggregation, forecasting, or regional comparison tooling.	7. BEHAVIOUR BE Planners manually download & open CSV/Excel exports each morning, cross-reference sheets, and call colleagues in other regions to check load status.	
Focus on J&P, understand RC	3. TRIGGERS TR Heatwave or festival season approaching; leadership requests a same-day briefing; sudden usage spike; renewable budget planning cycle.	10. YOUR SOLUTION SL A web-based electricity consumption analytics platform that automatically ingests, cleans, and aggregates state-wise usage; shows real-time regional dashboards and short-term demand forecasts; and sends threshold-based alerts — replacing manual spreadsheet work with one live, trusted view.	8.1 ONLINE CH Existing web/Excel reporting portals, internal email bulletins, BI dashboards, state utility websites.	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM Before: anxious, reactive, overwhelmed by scattered sheets. After: confident, in control, proactive.		8.2 OFFLINE Phone calls between regional control rooms, printed daily briefing sheets, in-person planning meetings.	
Identify strong TR & EM				